Project 2: Where is it?

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You are a LocatorBot. When robots accidentally get their internal maps of the ship wiped by cosmic rays, etc and so forth, it is your job to go out and find them and reboot their internal navigation. (Your own internal navigation is hardened against cosmic radiation, but that's an expensive job to do for all bots. You're worth it, though.)

You have at your disposal a bot detector (it goes ping when there's a bot). Using the feedback it gives you, you'll try to navigate your way to the bot as efficiently as possible.

1 The Task

For the purpose of this assignment, I want you to consider two models of bot detector. Type A: it delivers a ping if the bot is within a $(2k+1)$x$(2k+1)$ square (centered on the LocatorBot, a square of 'radius' k). If the bot is outside that radius, there is no ping.

Type B: this delivers a ping with high probability if the bot is nearby, and with lower probability the further away the bot is. The intutition here is if you run the detector multiple times and get no ping, the bot must be far away; if you run the detector multiple times and get a ping every time, the bot must be close by. The general formula is for a fixed $\alpha > 0$, and a, b different cells in the ship:

$$P(\text{ping}|\text{LocatorBot in Cell a, LostBot in Cell b}) = e^{-\alpha(\text{dist}(a,b)-1} \tag{1}$$

For this assignment, take the distance to be the euclidean distance between two cells.

> Question for analysis: What is the range of k you will want to consider here? What is the range of α we should consider here? Note, k and α specify the sensitivity of the detector - how/why?

If you are currently in the cell where LostBot is, and you run the detector, instead of a ping or no ping, the detector will tell you you've successfully located the bot, and the task ends.

2 Baseline Strategies

As before, I'll offer you a baseline strategy (or two) and your job will be to try to improve upon the performance of the baseline. Note, I'm not asking you to assert new abilities for the bots - I am simply ask you to consider *how can you make the best decision possible, given the information you have available to you?*

> **Baseline Strategy Type A:** Maintain a framework of where the LostBot may be.
>
> - Run the detector.
> - Update the possible locations of LostBot based on the results (ping vs no ping).
> - Move to the nearest cell that might contain LostBot (breaking ties at random).
> - Repeat until LostBot becomes FoundBot.

Baseline Strategy Type B: Maintain a framework of where the LostBot might be (i.e., a probabilistic knowledge base).

- Run the detector.
- Given the results of the detector (ping vs no ping), update your knowledgebase of where LostBot is.
- Move to the cell that is most likely to contain LostBot (breaking ties by distance, then at random).
- Repeat until LostBot becomes FoundBot.

3 Your Strategies

For both types of detector, your task is to come up with a strategy for discovering LostBot more efficient than the baseline strategies. For the purpose of this assignment, let us assume that running the detector is half the cost of moving to a nearby cell, so the total cost will be (number of move actions) + (1/2)(number of detector actions).

4 Data, Analysis, and Writeup

1) Give an explicit update rule for the knowledge bases for Type A Bot and Type B Bot, *and justify that your formulas are correct.*

2) Outline your strategy for using the Type A and Type B detectors. Be clear about any decisions you're making, any justifications you are using for your design choices. Why is your strategy a good idea, and why should it be better than the baseline choices? Try to utilize the ideas of the course to justify yourself (theoretically and/or practically). Be sure to specify how each bot strategy utilizes its current state of knowledge. Again, you should communicate your ideas in a way that leaves no question to the reader as to what your strategies are actually doing.

3) Compare Baseline Type A strategy to your Type A strategy, graphing the discovery cost as a function of k. Note you'll need to repeat multiple experiments to collect sufficient data here. Where is the most interesting k-regime? *Note, you should aim to do these experiments on as large a ship as you can, within the constraints of time allowed.*

4) Compare Baseline Type B strategy to your Type B strategy, graphing the discovery cost as a function of α.

   Note, be very careful here about what a good range of α is! How did you make your choices?

5) At the 'most efficient k' and the 'most efficient α', which detector type is more 'efficient', and *why*?

6) Suppose that movement is free, and the only thing that costs anything is running the detector. How does this change your strategies? Implement this change and compare your results to the baseline strategies and your previous strategies. Which detector is more efficient here? (Tuned optimally). Be clear in your analysis and experimentation.